

CHITRADURGA, India

WMO: 432330

Lat: 14.233N

Lon: 76.433E

Elev: 733

StdP: 92.82

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	15.7	16.9	1.4	4.6	29.2	4.0	5.5	28.4	5.8	24.3	5.2	24.0	0.5	90	0.270

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	11.6	36.7	20.8	35.7	20.8	34.8	20.9	25.5	31.0	24.7	30.0	24.2	29.2	1.8	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.1	20.8	28.4	23.4	19.9	27.4	22.9	19.3	26.8	83.1	31.1	79.9	30.0	77.6	29.2	29.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.6	5.6	4.9	DB	12.4	37.9	1.5	1.3	11.3	38.9	10.4	39.6	9.6	40.3	8.5	41.2	(4)
(5)			WB	9.5	26.7	1.1	1.2	8.7	27.5	8.0	28.2	7.4	28.8	6.6	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	25.6	23.5	25.9	28.6	29.5	28.5	25.8	24.6	24.4	24.8	24.9	23.6	22.9	(6)
(7)		DBStd	2.50	1.48	1.62	1.46	1.42	1.58	1.48	1.17	1.08	1.06	1.24	1.44	1.34	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	5684	418	447	575	586	572	475	453	447	443	462	408	399	(10)
(11)		CDD18.3	2642	159	213	317	336	314	225	194	189	193	203	158	141	(11)
(12)		CDH23.3	22513	1205	2102	3828	4236	3518	1558	903	836	1083	1363	995	887	(12)
(13)		CDH26.7	9021	343	926	1997	2288	1715	469	166	133	231	376	206	172	(13)
(14)	Wind	WSAvg	2.1	1.4	1.5	1.5	1.8	2.6	3.4	3.3	3.0	2.3	1.3	1.3	1.5	(14)
(15)	Precipitation	PrecAvg	607	0	1	4	45	76	53	78	70	110	123	40	12	(15)
(16)		PrecMax	992	4	18	33	144	174	157	206	157	301	283	134	56	(16)
(17)		PrecMin	377	0	0	0	0	4	6	20	17	15	0	0	0	(17)
(18)		PrecStd	149	1	4	8	32	45	33	46	41	67	66	40	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.9	34.9	37.1	38.3	37.3	33.7	30.8	30.2	31.3	32.0	30.5	30.7	(19)
(20)			MCWB	18.8	18.7	19.5	21.0	21.8	23.4	22.7	23.3	22.6	20.7	20.0	18.7	(20)
(21)		2%	DB	30.6	33.6	36.0	37.1	35.9	31.9	29.6	29.2	30.1	30.7	29.6	29.4	(21)
(22)			MCWB	18.5	19.0	19.7	20.7	22.2	23.2	22.6	23.1	22.5	20.7	20.1	18.8	(22)
(23)		5%	DB	29.6	32.6	35.1	36.1	34.8	30.6	28.5	28.2	29.1	29.9	28.8	28.4	(23)
(24)			MCWB	18.3	18.7	19.7	20.9	22.4	23.1	22.6	22.8	22.4	20.9	20.1	18.7	(24)
(25)		10%	DB	28.7	31.5	34.3	35.1	33.6	29.4	27.4	27.2	28.1	28.9	27.9	27.5	(25)
(26)			MCWB	18.1	18.5	19.7	21.2	22.7	23.0	22.5	22.6	22.3	21.4	20.0	18.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.3	23.5	25.9	27.2	26.1	25.6	24.6	24.7	24.7	24.5	23.6	22.7	(27)
(28)			MCDWB	27.4	30.9	33.6	33.5	31.6	29.7	28.7	28.5	28.2	27.9	27.2	27.2	(28)
(29)		2%	WB	21.2	22.1	24.2	25.6	25.1	24.6	23.7	23.7	24.0	23.7	22.8	21.7	(29)
(30)			MCDWB	26.7	28.9	30.6	31.7	30.8	29.1	27.3	27.3	27.7	27.3	26.4	25.8	(30)
(31)		5%	WB	20.5	21.2	23.2	24.6	24.6	24.0	23.2	23.2	23.5	23.2	22.2	21.1	(31)
(32)			MCDWB	26.3	28.0	29.8	30.3	30.2	28.5	26.6	26.6	27.1	26.9	25.9	25.1	(32)
(33)		10%	WB	19.7	20.5	22.4	23.8	24.1	23.5	22.7	22.8	23.0	22.7	21.7	20.5	(33)
(34)			MCDWB	25.8	27.3	28.8	29.3	29.6	27.8	25.9	26.0	26.5	26.4	25.3	24.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.8	11.3	11.6	11.6	10.3	7.0	5.7	5.7	6.7	7.9	8.8	9.9	(35)
(36)			MCDBR	11.7	12.2	12.4	12.5	11.7	8.9	7.6	7.2	8.3	9.8	10.1	11.1	(36)
(37)			MCWBR	4.9	5.4	5.5	5.3	3.8	2.9	2.5	2.7	2.9	3.5	4.1	4.6	(37)
(38)		5% WB	MCDWB	10.3	11.1	11.3	11.8	10.4	7.6	6.1	6.2	7.0	7.3	7.4	8.8	(38)
(39)			MCWBR	4.9	5.5	5.5	5.3	4.1	3.2	2.7	2.8	3.3	3.3	3.5	4.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.392	0.414	0.481	0.530	0.535	0.479	0.489	0.471	0.449	0.485	0.433	0.409	(40)	
(41)		taud	2.261	2.212	2.037	1.949	1.964	2.176	2.141	2.204	2.266	2.127	2.237	2.251	(41)	
(42)		Ebn at Noon	900	895	842	797	780	816	810	834	857	818	853	872	(42)	
(43)		Edh at Noon	132	145	177	193	187	149	155	147	138	156	135	131	(43)	
(44)	All-Sky Solar Radiation	RadAvg	5.42	6.17	6.60	6.72	6.42	5.12	4.48	4.61	5.04	5.07	4.87	4.96	(44)	
(45)		RadStd	0.22	0.24	0.23	0.18	0.28	0.38	0.30	0.22	0.35	0.35	0.36	0.25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)	
(47) Regional (4 neighbors)		+0.32	N/A	N/A	+0.52	N/A	N/A	N/A	N/A	+171	+172	(47)	

Nomenclature: See separate page